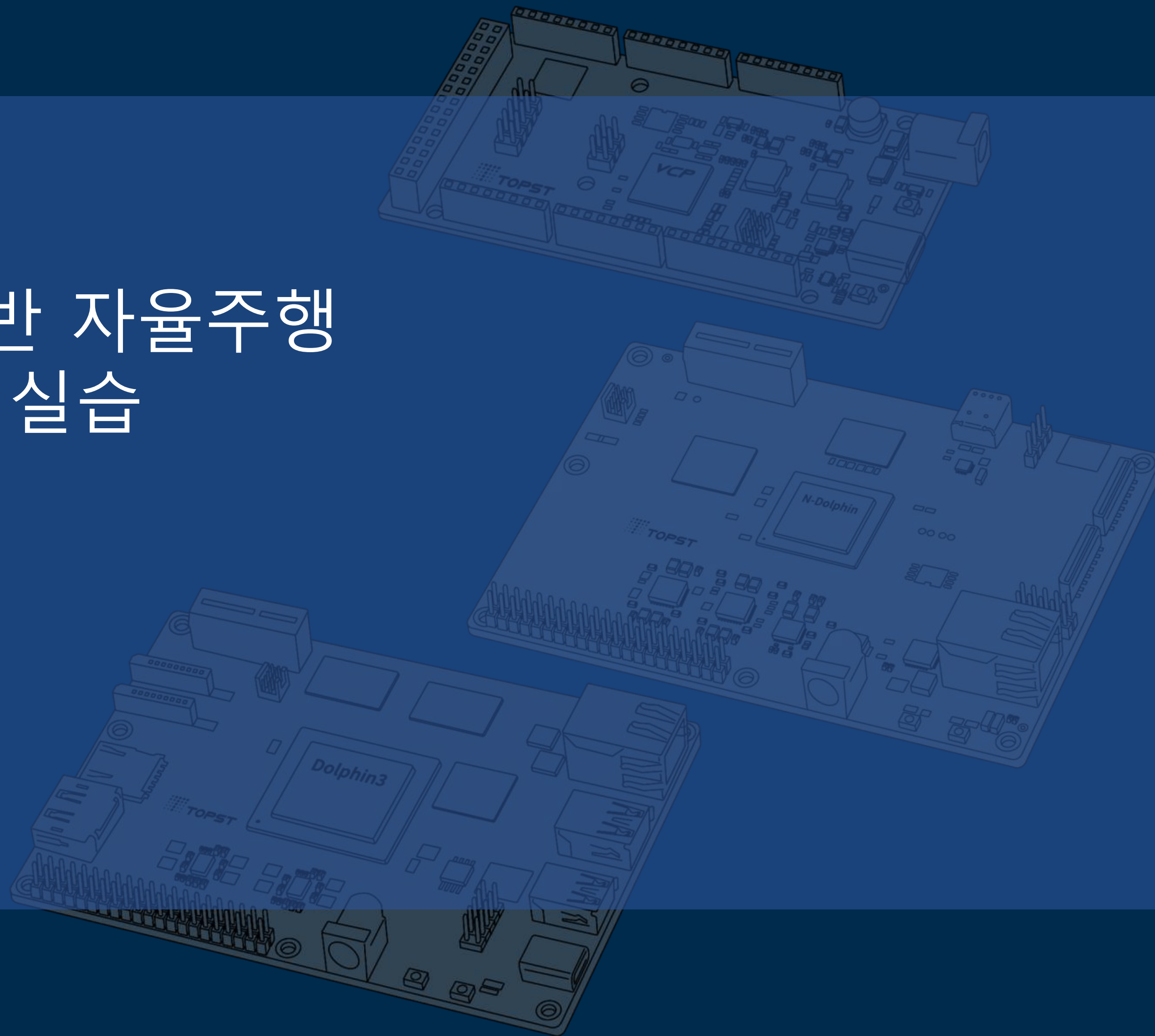
팹리스 실무형 일경험 프로그램

SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습



6일차

01 AI-G 환경 구성

Telechips TOPST

목차

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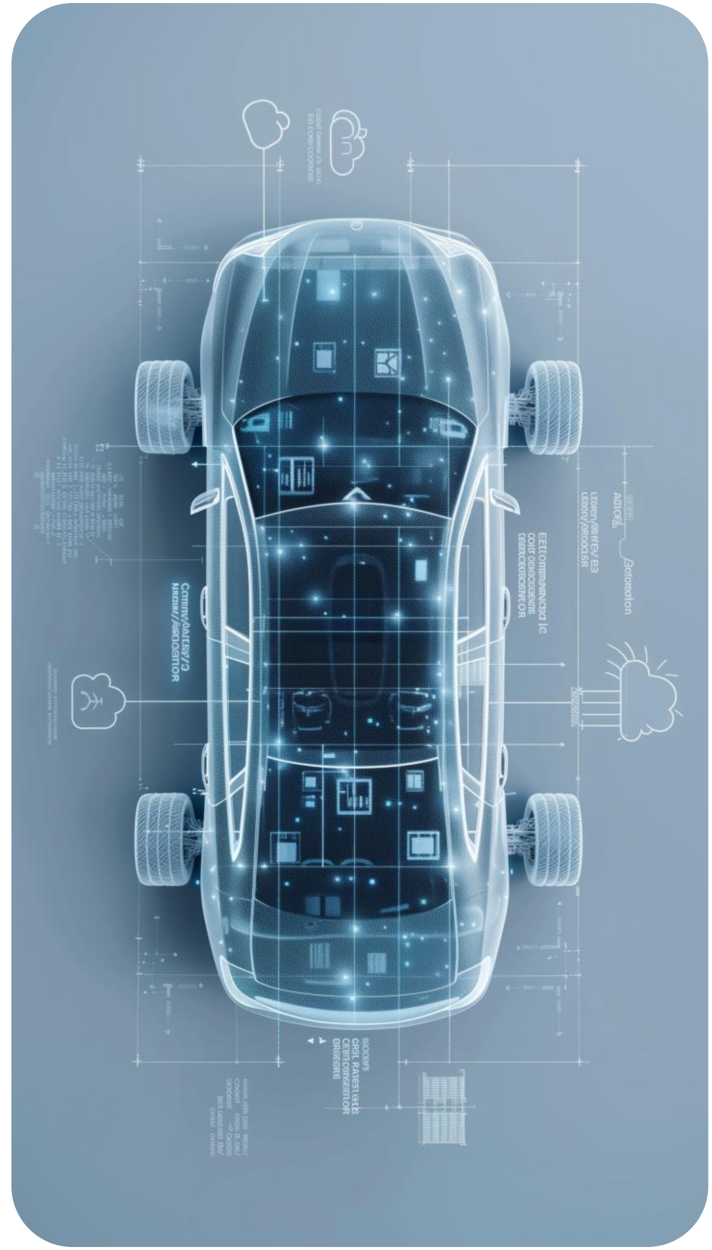


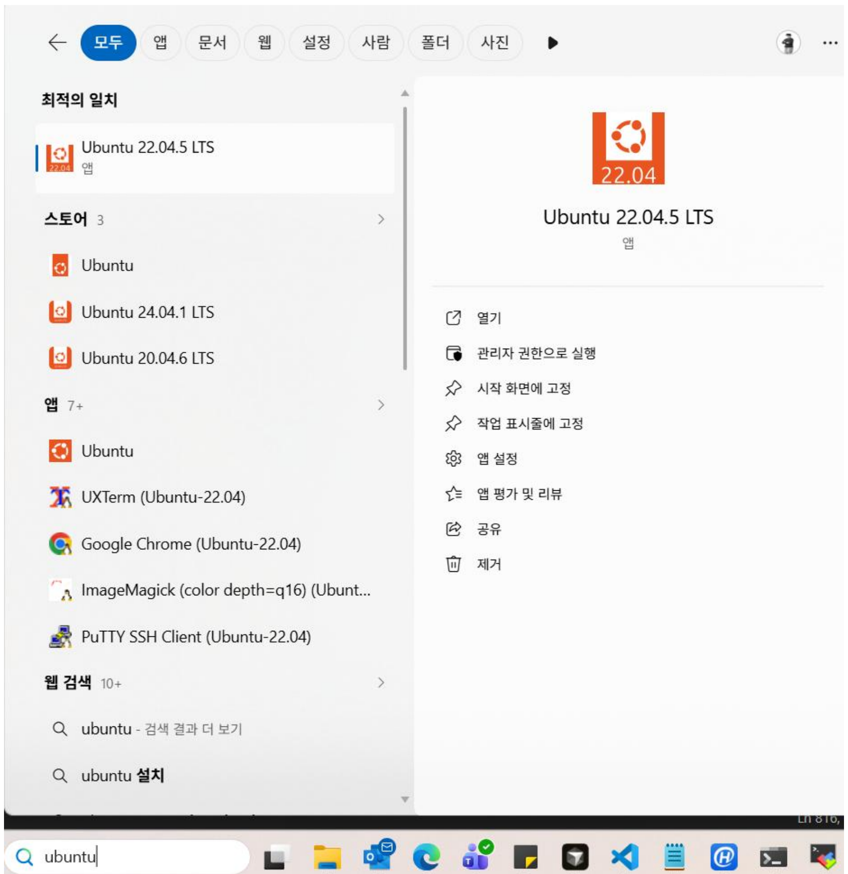
AI-G Specification

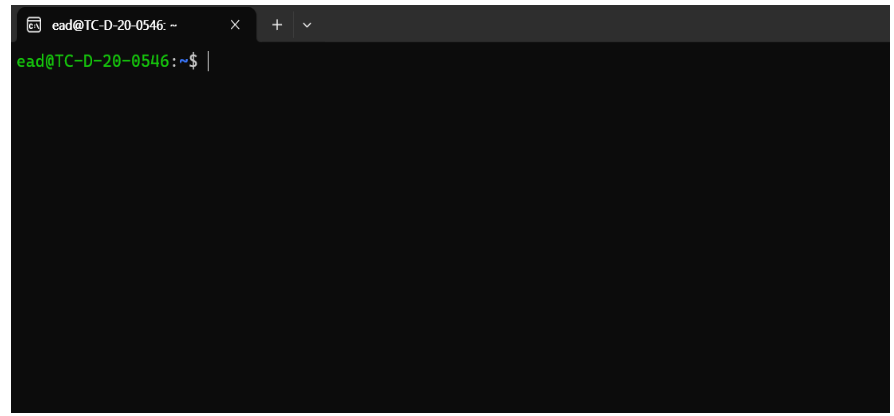
Part Name	AI-G
CPU	Arm® Cortex-A53 1.45 GHz, Quad
NPU	1.0 Ghz, 8 TOPS
Memory	2 GB 32-bit LPDDR4X
Storage	8 GB eMMC 5.1 Flash
Network	Gigabit Ethernet
Camera	MIPI CSI-2 2-lane, MIPI CSI-2 4-lane
Display	MIPI DSI-2 2-lane
Button	System reset button, Boot mode button
Others	PCIe Gen3, FWD debugger port, I2S, I2C, SPI,CAN, JTAG
Power	5V/3A
OS	Linux
Mechanical	120 mm x 90 mm

- AI-G 는 1개의 CPU와 1개의 NPU 존재
  - Cortex-A53
  - NPU : 8TOPS (4TOPS Dual)
  - 2GB RAM
  - 8GB eMMC
  - Ethernet, DSI, CSI, PCIe,

AI-G Yocto

- **WSL Ubuntu 22.04 실행**





AI-G Yocto

Build Source 가져오기 (WSL Ubuntu)

$ cd ~/Work

$ mkdir ai-g-yocto

$ cd ai-g-yocto

$ repo init -u https://github.com/topst-development/manifests.git -b develop -m linux_yp4.0_topst.xml

```
ead@TC-D-20-0546:~/topst-sdk$ repo init -u https://github.com/topst-development/manifests.git -b develop -m
 linux_yp4.0_topst.xml
Downloading Repo source from https://gerrit.googlesource.com/git-repo

... A new version of repo (2.54) is available.
... New version is available at: /home/ead/topst-sdk/.repo/repo/repo
... The launcher is run from: /usr/bin/repo
!!! The launcher is not writable.  Please talk to your sysadmin or distro
!!! to get an update installed.


Your identity is: Topst Developer <topstdeveloper@gmail.com>
If you want to change this, please re-run 'repo init' with --config-name

repo has been initialized in /home/ead/topst-sdk
```

AI-G Yocto

```
$ repo sync
```

```
ead@TC-D-20-0546:~/topst-sdk$ repo sync

... A new version of repo (2.54) is available.
... New version is available at: /home/ead/topst-sdk/.repo/repo/repo
... The launcher is run from: /usr/bin/repo
!!! The launcher is not writable.  Please talk to your sysadmin or distro
!!! to get an update installed.

Syncing:  25% (3/12), done in 35.713s
Syncing: 100% (12/12) 0:52 | 1 job | 0:14 meta-openembedded @ poky/meta-openembeddedrepo sync has finished
successfully.
```

```
$ ls
```

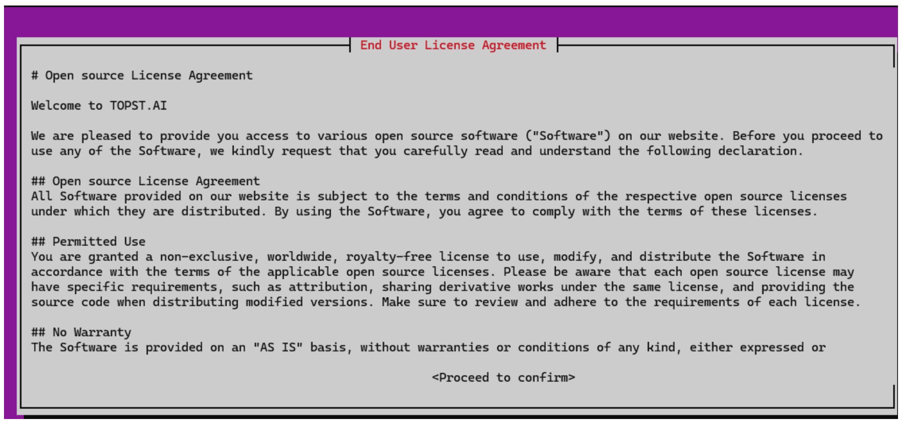
```
ead@TC-D-20-0546:~/topst-sdk$ ls
easy-setup.sh  mktcimg  poky  stitch-fai-ai.sh  stitch-fai-d3.sh  tools
```

AI-G Yocto

topst-build.sh 실행

```
$ ./easy-setup.sh
```

스크롤 내리기 -> 방향키(->) 클릭 -> 엔터 -> accept 엔터



Telechips

AI-G Yocto

topst-build.sh 실행

$ source poky/meta-topst/topst-build.sh -> 1번 머신 선택 (1 입력)

```
ead@TC-D-20-0546:~/topst-sdk$ source poky/meta-topst/topst-build.sh
Choose MACHINE
  1. ai-g-topst
  2. d3-g-topst-main
  3. d3-g-topst-sub
  4. d5-g-topst-main
  5. d5-g-topst-sub
select number(1-5) => 1
machine(ai-g-topst) selected.
You had no conf/local.conf file. This configuration file has therefore been
created for you from /home/ead/topst-sdk/poky/meta-topst/template/ai-g-topst/local.conf.sample
You may wish to edit it to, for example, select a different MACHINE (target
hardware). See conf/local.conf for more information as common configuration
options are commented.
```

```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst$ ls
bitbake-cookerdaemon.log  buildhistory  cache  conf  downloads  sstate-cache  tmp
```

AI-G Yocto

쓰레드 개수 조정하기

현재는 쓰레드 수가 8로 되어 있음. 개인 pc의 스펙을 고려해 각 태스크 수와 태스크 내부의 job수 를 4로 수정

```
$ vi conf/local.conf
```

```
# variable as required.

# Parallelism Options
#
# These two options control how much para
# option determines how many tasks bitbak
#
BB_NUMBER_THREADS ?= "8"
#
# Default to setting automatically based
#BB_NUMBER_THREADS ?= "${@bb.utils.cpu_co
#
# The second option controls how many pro
# running compile tasks:
#
PARALLEL_MAKE ?= "-j 16"
```

AI-G Yocto

쓰레드 개수 조정하기

현재는 쓰레드 수가 8로 되어 있음. 개인 pc의 스펙을 고려해 각 태스크 수와 태스크 내부의 job수 를 4로 수정

```
$ vi conf/local.conf
```

```
# Parallelism Options
#
# These two options control how muc
# option determines how many tasks
#
BB_NUMBER_THREADS ?= "4"
#
# Default to setting automatically
#BB_NUMBER_THREADS ?= "${@bb.utils.
#
# The second option controls how ma
# running compile tasks:
#
PARALLEL_MAKE ?= "-j 4"
```

AI-G Yocto

저장

- 수정 후 저장하는 법 - esc -> : 입력-> wq 입력 -> 엔터

AI-G tc-nn-app 빌드

tc-nn-app 빌드 수행

- tc-nn-app은 AI-G의 NPU를 사용하기 위한 추론 용 어플리케이션
- 카메라나 이미지를 입력으로 받아 추론 결과를 디스플레이나 이미지 형태로 제공
- 이후 강의에서 tc-nn-app을 통해 차선 인식을 수행할 예정
- 기존의 tc-nn-app에서는 차선 인식 기능이 추가되어 있지 않아 앱 빌드 후 코드 수정 필요

AI-G tc-nn-app 빌드

tc-nn-app 빌드 수행

$ bitbake -c compile -f tc-nn-app

```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst$ bitbake -c compile -f tc-nn-app
Loading cache: 100% |##########################################################################################| Time: 0:00:00
Loaded 3867 entries from dependency cache.
Parsing recipes: 100% |########################################################################################| Time: 0:00:01
Parsing of 2473 .bb files complete (2460 cached, 13 parsed). 3881 targets, 318 skipped, 0 masked, 0 errors.
WARNING: No recipes in default available for:
  /home/ead/topst-sdk/poky/meta-topst/recipes-browser/cog/cog_0.16.%.bbappend
NOTE: Resolving any missing task queue dependencies

Build Configuration:
BB_VERSION           = "2.0.0"
BUILD_SYS            = "x86_64-linux"
NATIVELSBSTRING      = "universal"
TARGET_SYS           = "aarch64-telechips-linux"
```

```
Initialising tasks: 100% |#####################################################################################| Time: 0:00:02
Checking sstate mirror object availability: 100% |#############################################################| Time: 0:00:04
Sstate summary: Wanted 506 Local 164 Mirrors 0 Missed 342 Current 1017 (32% match, 77% complete)
NOTE: Executing Tasks
NOTE: Tasks Summary: Attempted 3899 tasks of which 3226 didn't need to be rerun and all succeeded.
NOTE: Writing buildhistory
NOTE: Writing buildhistory took: 3 seconds

Summary: There was 1 WARNING message.
```

AI-G tc-nn-app 빌드

Yocto 빌드 결과 dir 설명

- Kernel : ai-g-topst-tc-boot-5.10.223-r0.img, Image--5.10.223-r0-ai-g-topst-20251030223228.bin
- Kernel Modules : modules--5.10.223-r0-ai-g-topst-20251030223228.tgz
- Device tree bin : tcc7500-lpd4x321--5.10.223-r0-ai-g-topst-20251030223228.dtb
- Rootfs : telechips-topst-image-ai-g-topst-20251030223228.rootfs.ext4

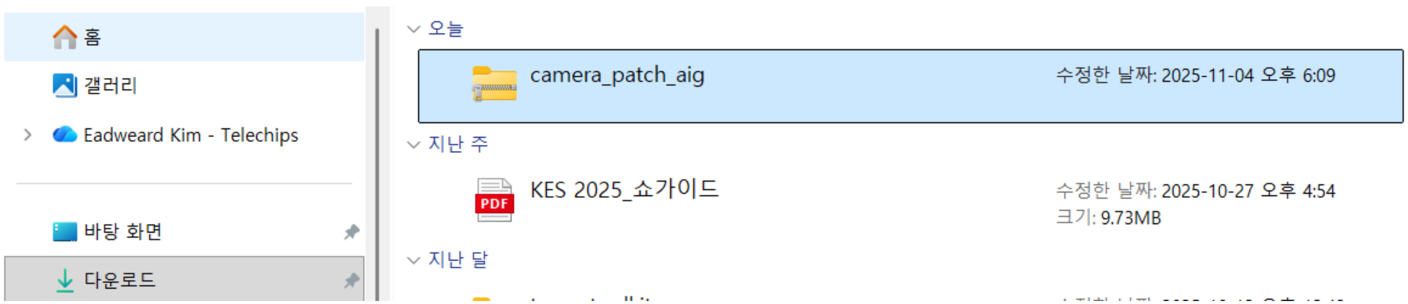
```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst/tmp/deploy/fwdn$ ls
boot-firmware  partition.list  SD_Data.fai  SD_Data.gpt  SD_Data.gpt.back  SD_Data.gpt.prim
```

```
ead@TC-D-20-0546:~/topst-sdk/build/ai-g-topst/tmp/deploy/images/ai-g-topst$ ls
ai-g-topst-tc-boot-5.10.223-r0.img                             tc-boot.img
boot-firmware                                                  tcc7500-lpd4x321--5.10.223-r0-ai-g-topst-20251030223228.dtb
Image                                                          tcc7500-lpd4x321-ai-g-topst.dtb
Image--5.10.223-r0-ai-g-topst-20251030223228.bin               tcc7500-lpd4x321.dtb
Image-ai-g-topst.bin                                           telechips-topst-ai-image-ai-g-topst-20251030223228.rootfs.ext4
Image-initramfs--5.10.223-r0-ai-g-topst-20251030223228.bin     telechips-topst-ai-image-ai-g-topst-20251030223228.rootfs.manifest
Image-initramfs-ai-g-topst.bin                                 telechips-topst-ai-image-ai-g-topst-20251030223228.testdata.json
initramfs-topst-image-ai-g-topst-20251030223228.rootfs.cpio    telechips-topst-ai-image-ai-g-topst.ext4
initramfs-topst-image-ai-g-topst-20251030223228.rootfs.manifest telechips-topst-ai-image-ai-g-topst.manifest
initramfs-topst-image-ai-g-topst-20251030223228.testdata.json  telechips-topst-ai-image-ai-g-topst.testdata.json
initramfs-topst-image-ai-g-topst.cpio                          u-boot-ai-g-topst-1.0-r0.rom
initramfs-topst-image-ai-g-topst.manifest                      u-boot-ai-g-topst.rom
initramfs-topst-image-ai-g-topst.testdata.json                 u-boot.rom
modules--5.10.223-r0-ai-g-topst-20251030223228.tgz             u-boot-topst-initial-env
modules-ai-g-topst.tgz                                         u-boot-topst-initial-env-ai-g-topst
tc-boot-ai-g-topst.img                                         u-boot-topst-initial-env-ai-g-topst-1.0-r0
```

AI-G 환경 설정

AI-G 이미지 다운 : AI-G Yocto image

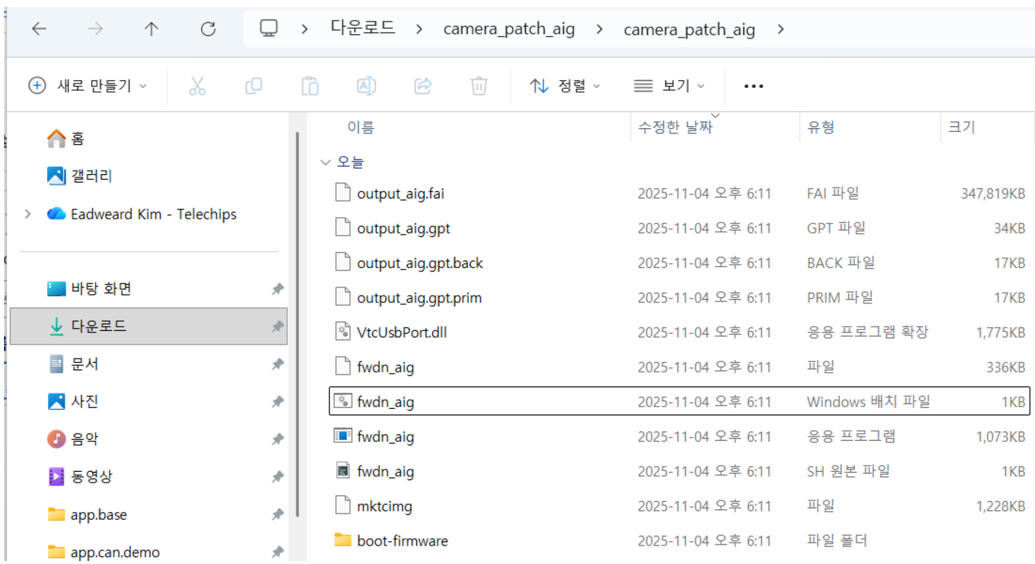
설치 후 다운로드 탭에서 해당 파일 압축 해제



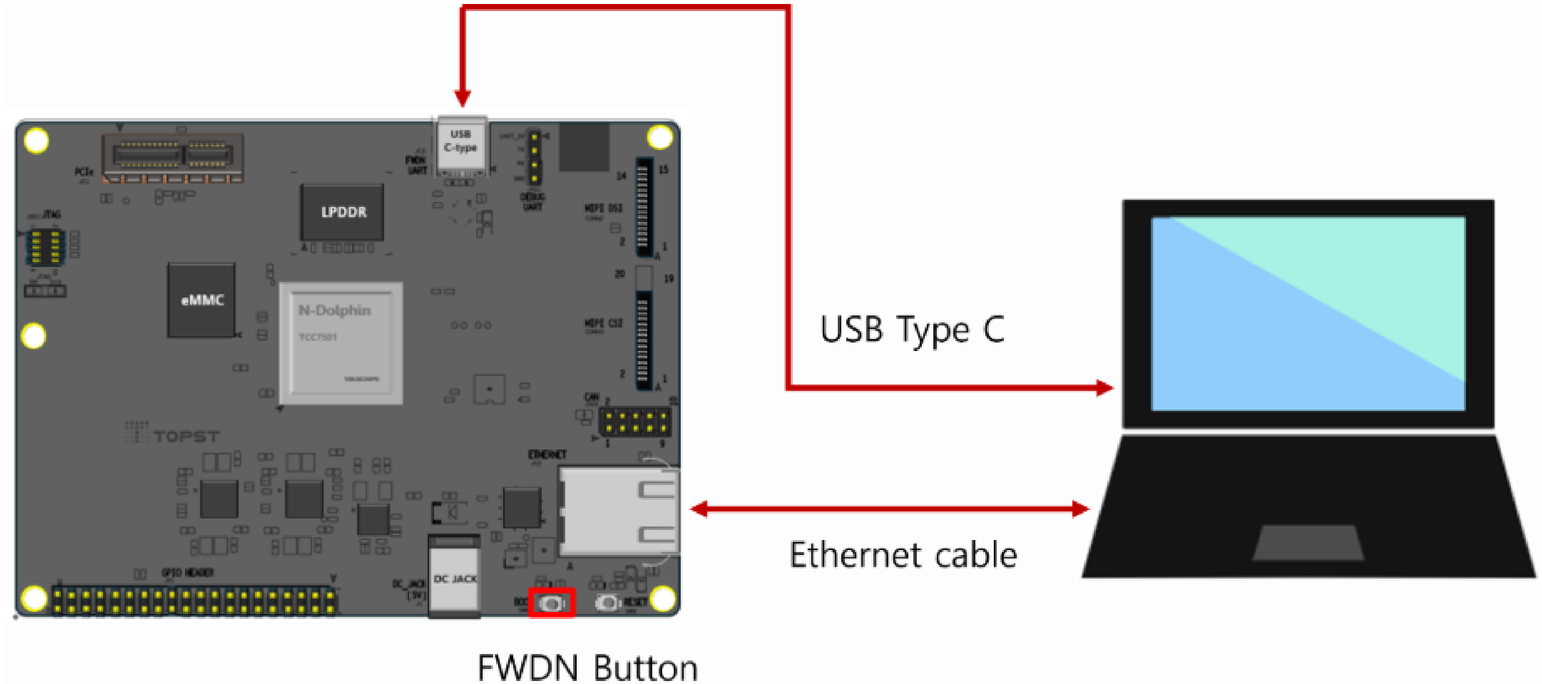
AI-G 환경 설정

AI-G 이미지 다운 : AI-G Yocto image

`fwdn_aig` 배치파일 존재하는지 확인

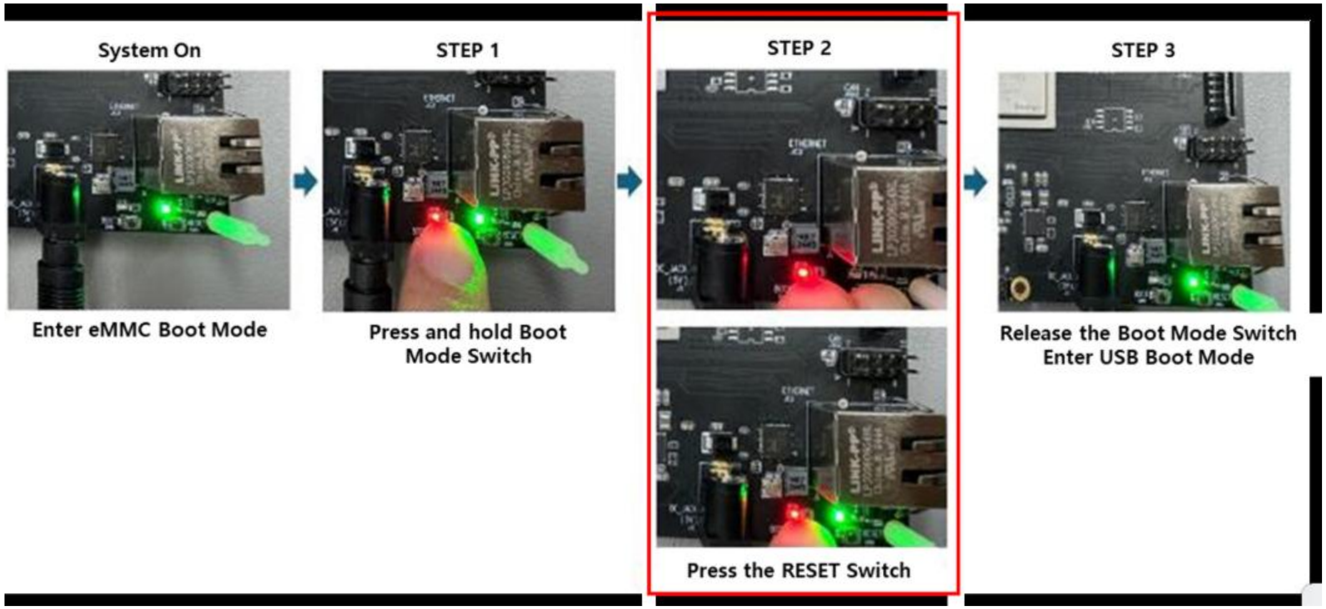


AI-G 환경 설정



- **Connect AI-G Board and Host PC with USB Boot Mode**
  - VTC 드라이버가 PC에 정상적으로 설치되어 있는지 확인
  - USB Type-C 케이블을 Host PC 및 AI-G의 USB Type-C FWDN 포트에 연결
  - 이더넷 케이블을 AI-G 보드의 이더넷 포트와 Host PC에 연결
  - FWDN 스위치를 누른 상태에서 전원 케이블을 AI-G 보드에 연결(D3-G보드와 동일)

AI-G 환경 설정



- **Connect AI-G Board and Host PC with USB Boot Mode**
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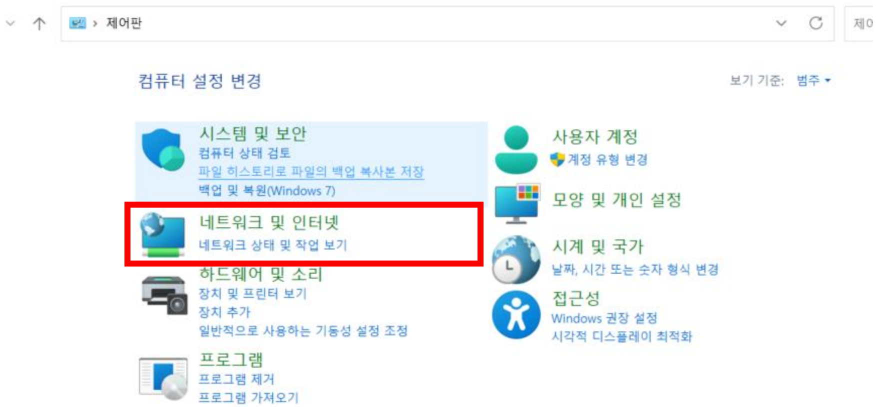
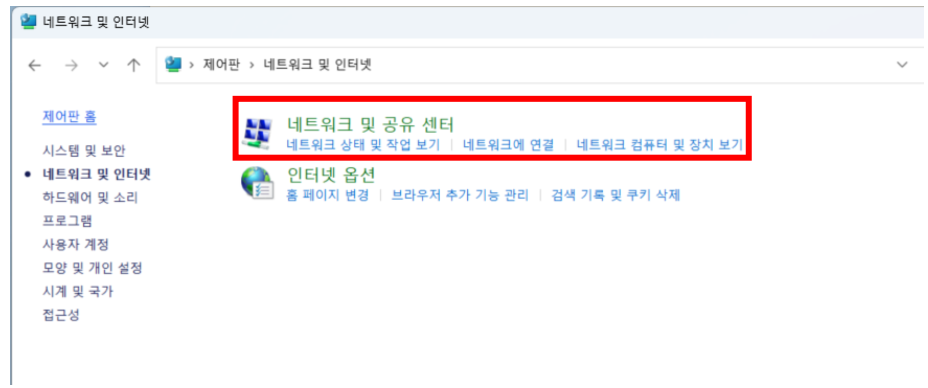
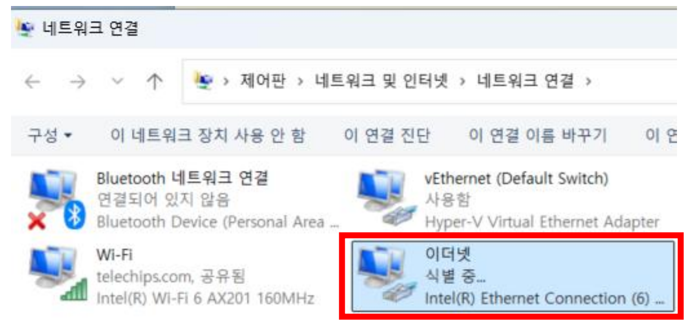
AI-G 환경 설정





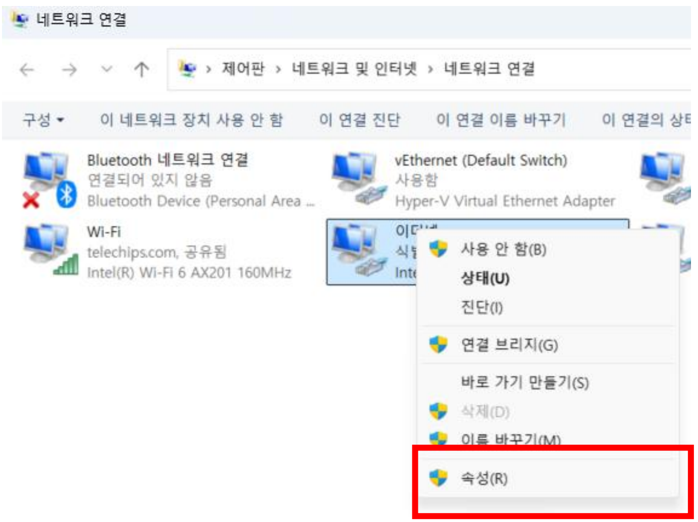
- **Ethernet Set Up**
  - Host PC Network Configuration
    - ➢ 제어판 -> 네트워크 및 인터넷 -> 네트워크 및 공유 센터 -> 어댑터 설정 변경

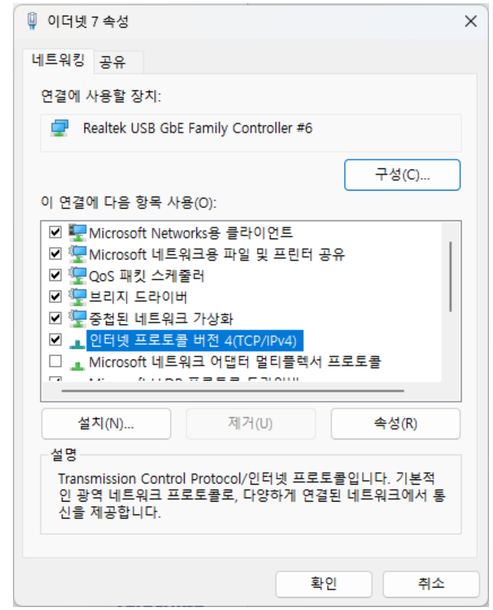
AI-G 환경 설정



- **Ethernet Set Up**
    - Host PC Network Configuration
        - 제어판 -> 네트워크 및 인터넷 -> 네트워크 및 공유 센터 -> 어댑터 설정 변경

AI-G 환경 설정





- **Ethernet Set Up**
  - Host PC Network Configuration
    - 이더넷 선택 -> 속성 -> 인터넷 프로토콜 버전4(TCP/IPv4) -> 속성

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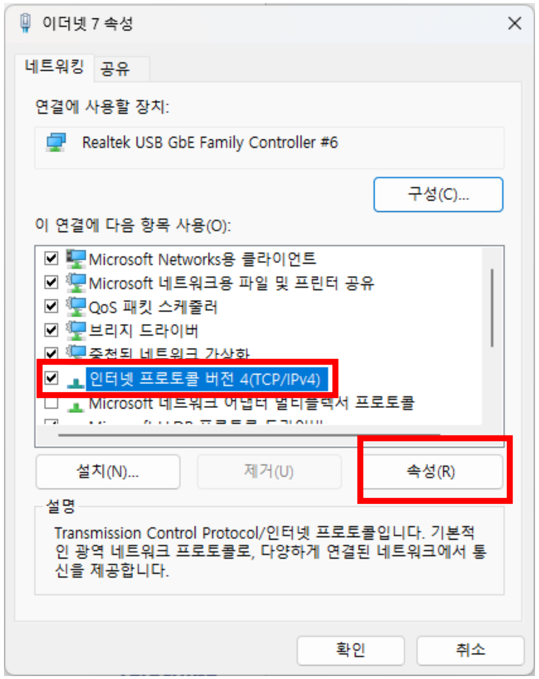
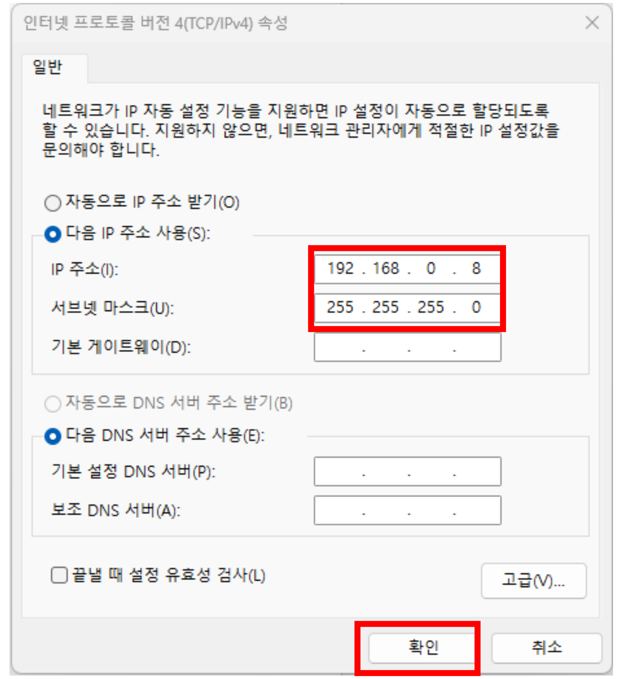
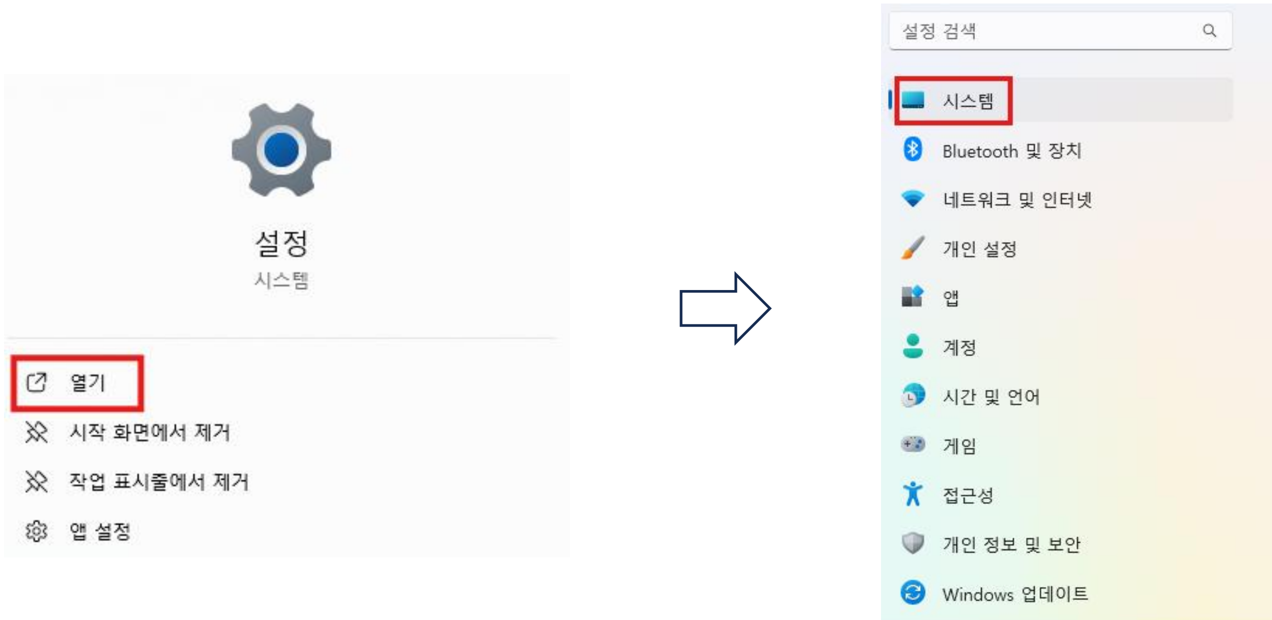
AI-G 환경 설정





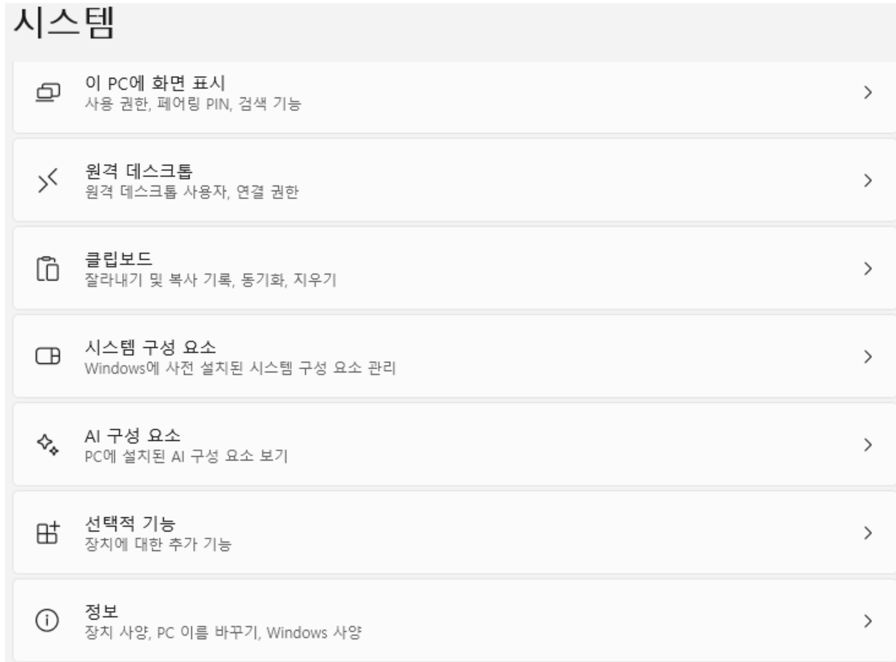
- **Ethernet Set Up**
  - Host PC Network Configuration
    - 다음 IP 주소 사용 -> IP 주소 : 192.168.0.8 / 서브넷 마스크 : 255.255.255.0 -> 확인

AI-G 환경 설정



- **Add WMIC (설치되어 있지 않은 경우 만)**
  - FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
    - 시작 메뉴에서 설정 열기
    - 시스템 메뉴로 이동

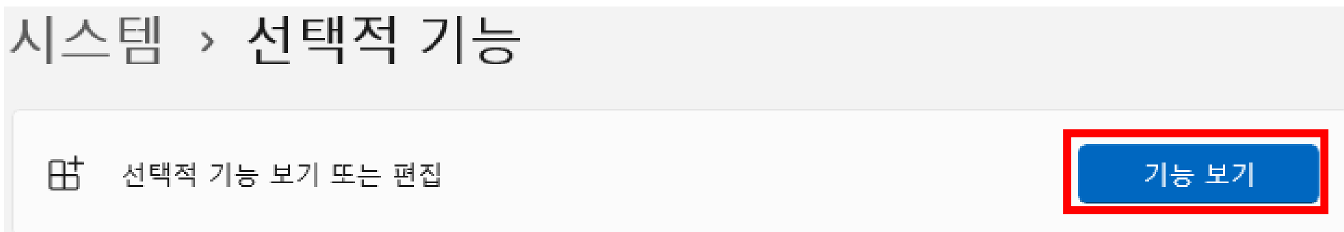
AI-G 환경 설정



- **Add WMIC (설치되어 있지 않은 경우 만)**
  - FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
    - ➢ 선택적 기능 메뉴 선택

AI-G 환경 설정

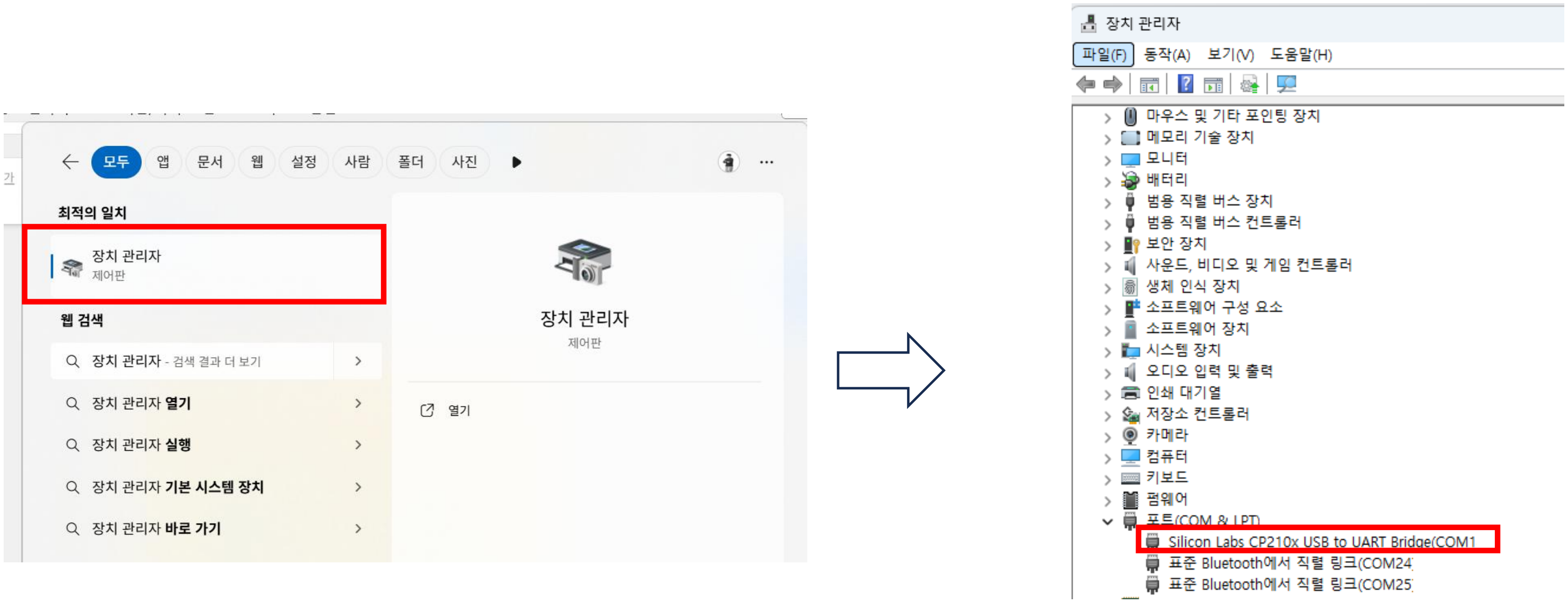
- ## Add WMIC
  - FWDN 전 AI-G 보드의 FWDN 포트에 연결된 포트를 알기 위해 WMIC 설치 필요
    - ➢ 기능 보기 선택 -> 사용 가능한 기능 보기



    - ➢ 검색창에 'WMIC' 입력

    - ➢ WMIC를 선택하여 다운로드

AI-G FWDN



- **FWDN (AI-G Yocto Image)**
  - 검색창에 장치 관리자 검색 후 실행
  - 포트 탭에서 연결된 장치 시리얼 번호 (COM xx 확인)

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AI-G FWDN

- **장치가 보이지 않을 경우**
  - 기타 장치에서 CP2102 우클릭 -> 드라이버 업데이트 -> 내 컴퓨터에서 드라이버 찾아보기
  - 찾아보기 -> 이전에 설치한 CP210x 드라이버 폴더 선택 후 확인 -> 다음
  - 장치 인식 확인

AI-G FWDN

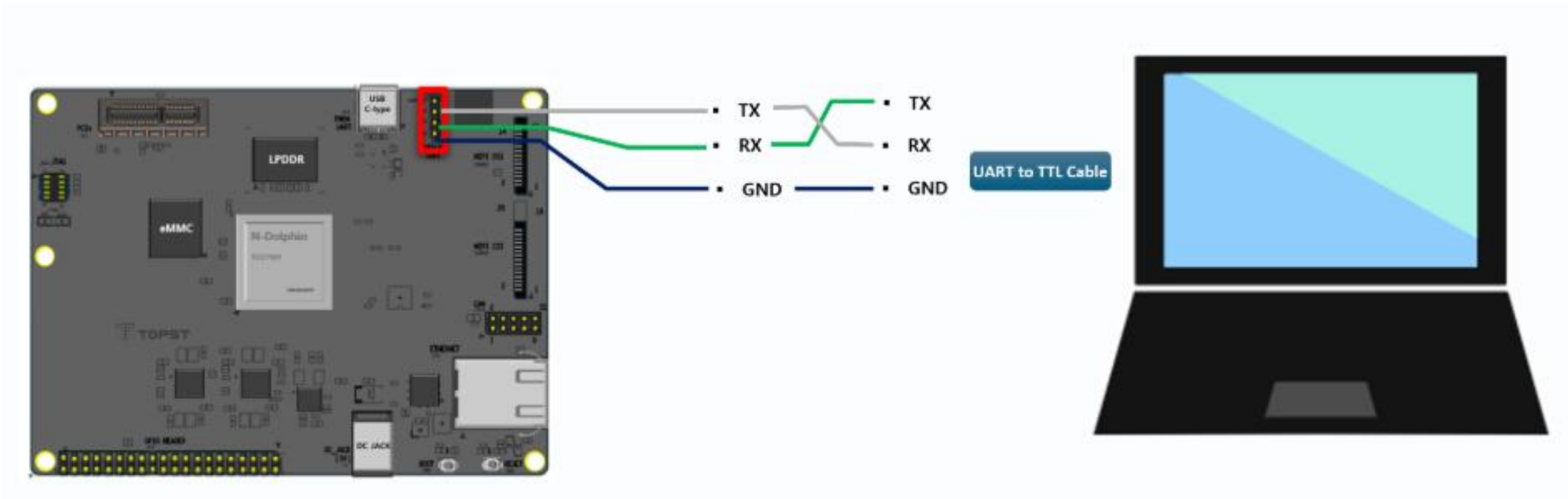
C:₩WINDOWS₩system32₩cm

```
TOPST AI-G FWDN Batch File

Find Port.
Silicon Labs CP210x USB to UART Bridge(COM7)
표준 Bluetooth에서 직렬 링크(COM24)
Bluetooth Device (RFCOMM Protocol TDI)
표준 Bluetooth에서 직렬 링크(COM25)

Input USB Port Number :
```

```
TOPST AI-G FWDN Batch File

Find Port.
Silicon Labs CP210x USB to UART Bridge(COM44)

Input USB Port Number : 44


Step 1. Connect between Host PC and TOPST AI-G
[FWDNLogger::PrintCurTime:100] 25/06/12-16:11:36
[main:24] FWDN N-Dolphin v1.0.1 - 2024.9.24 16:18:11
[main:92] Start FWDN
[FWDN_ND::ImageSend:442] Complete to send MCERT image
[FWDN_ND::ImageSend:442] Complete to send HSM image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL1 image
[FWDN_ND::ImageSend:442] Complete to send FWDN_DRAM_PARAMETER image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL2 image
[FWDN_ND::ImageSend:442] Complete to send FWDN_BL3 image
[FWDN_ND::InfoDevice:123] ############### Device Info ###############
[PrintIP:527] ip      : 192.168.0.100
[PrintIP:527] mask    : 255.255.255.0
[PrintIP:527] gateway : 0.0.0.0
[FWDN_ND::InfoDevice:132] port : 8080
[FWDN_ND::InfoDevice:135]
mmc0_user: 7818182656 byte block_size: 512 byte
mmc0_boot0: 4194304 byte block_size: 512 byte
mmc0_boot1: 4194304 byte block_size: 512 byte
snor1: 0 byte
snor2: 0 byte
fw_version: tcc750x_v0.0.111
fw_build_id: g90e2b777
fw_build_date: 2024-09-24-17:11:42+0900

----- Firmware Information -----
VERSION    : tcc750x_v0.0.111
BUILD ID   : g90e2b777
BUILD DATE : 2024-09-24-17:11:42+0900
```

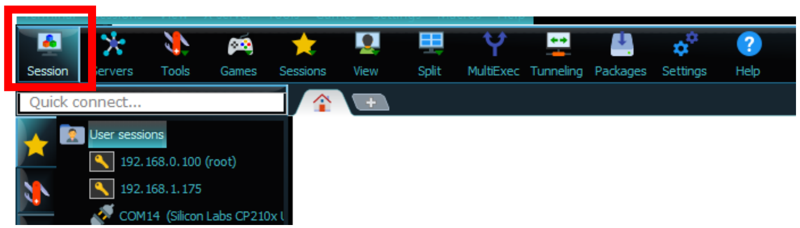
- **FWDN 실행**
  - fwdn_aig (windows 배치파일) 실행
  - USB 포트 넘버에 확인한 시리얼 장치 번호 입력 - 14

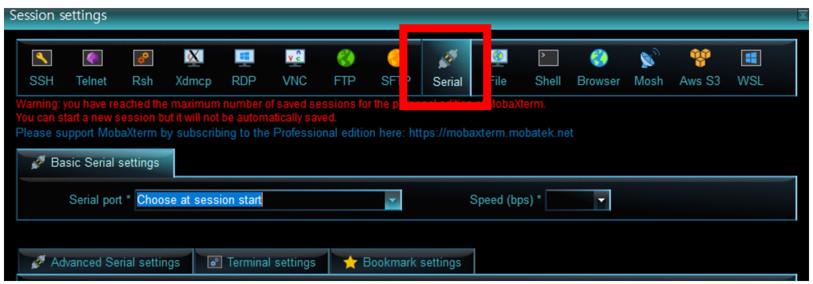
AI-G ↔ PC 통신



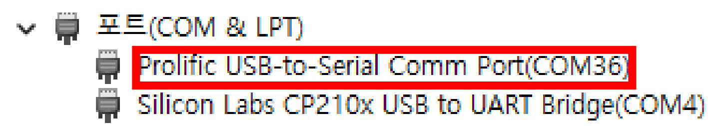
- **Connect AI-G Board and Host PC with UART**
  - AI-G 보드와 PC 간 USB-to-TTL cable을 이용하여 연결
    - ➢ 검정 케이블은 UART 핀의 GND에 연결
    - ➢ 흰 케이블은 UART 핀의 TX 핀에 연결 & 녹색 케이블은 UART 핀의 RX 핀에 연결

AI-G ↔ PC 통신





- **Connect AI-G Board and Host PC with UAR**
  - 보드 접속
    - ➢ MobaXterm 실행
    - ➢ 장치 관리자를 통해 port 확인



    - ➢ Session -> Serial 탭 클릭
    - ➢ Serial port 탭에서 장치관리자를 통해 확인한 포트 선택
    - ➢ Speed '115200'으로 설정

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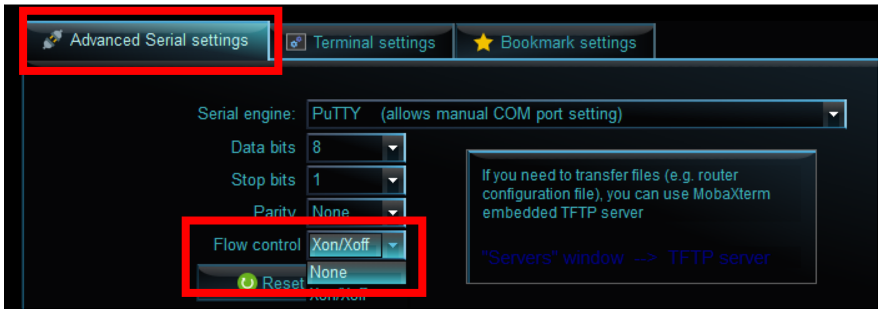
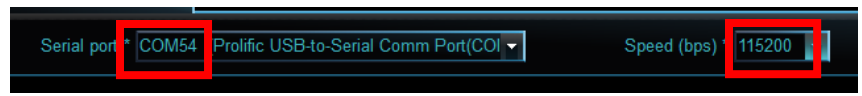
AI-G ↔ PC 통신

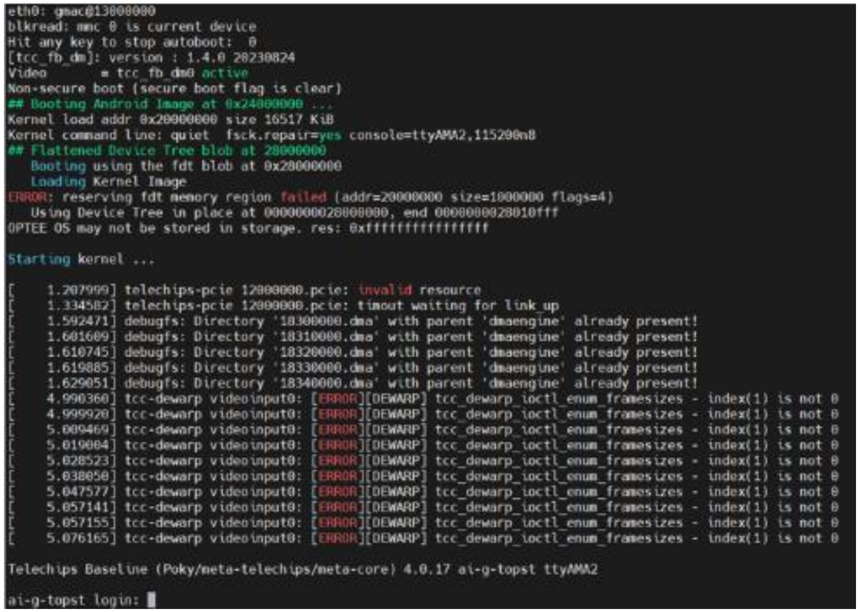


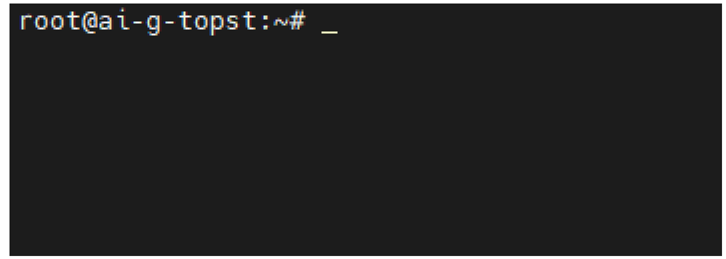
- Serial port에서 연결된 포트 설정
- Speed 115200 설정



- Advanced Serial settings 탭 선택
- Flow contro에서 None 선택
- OK 눌러 보드 접속

AI-G ↔ PC 통신





- 보드 실행
  - Reset 버튼을 누르거나 전원 재인가
  - 디버깅 창 확인
  - ai-g-topst login : root / password : root 입력

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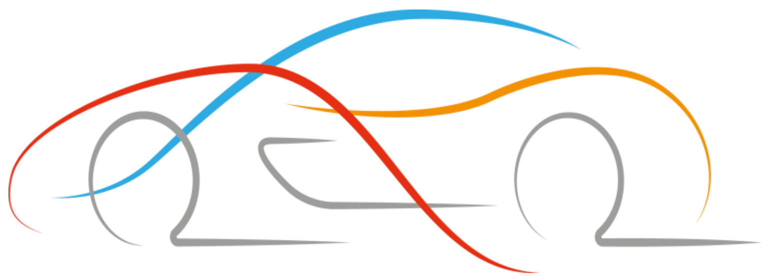

Thank You

Telechips